AELL101

Lecture Notes

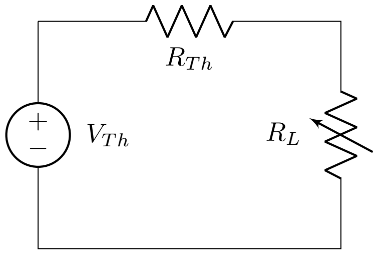
Group 5

January 28th - 30th

Network Theorems (continued)

Maximum Power Transfer

The maximum power is transferred to the load from a network when the load resistance (R_L) is equal to the Thevenin resistance (R_{Th}) seen from the load.



When maximum power is transferred to the load:

$$R_L = R_{Th}$$

The power transferred through the load is given by:

$$P_L = I^2 R_L$$

Substituting the current in terms of voltage:

$$P_L = \frac{V_{Th}^2}{(R_L + R_{Th})^2} R_L$$

When $R_L = R_{Th}$, the power becomes:

$$P_L = \frac{V_{Th}^2}{4R_{Th}}$$

The efficiency of the system is:

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{I^2 R_L}{I^2(R_L + R_{Th})} = \frac{R_L}{R_L + R_{Th}} = \frac{1}{1 + \frac{R_{Th}}{R_L}}$$

Thus, the efficiency when $R_L = R_{Th}$ is:

$$\eta = \frac{1}{2} \quad \text{or} \quad 50\%$$

Power Transfer vs. Efficiency

Although the theorem ensures maximum power delivery to the load, it does not guarantee maximum efficiency. The efficiency of power transfer increases when $R_L > R_{Th}$, as a greater proportion of power is delivered to the load. However, this also reduces the total power received by the load.
In contrast, if $R_L < R_{Th}$, the efficiency decreases because more power is dissipated within the Thevenin resistance. Although total power consumption increases, the power delivered to the load decreases.

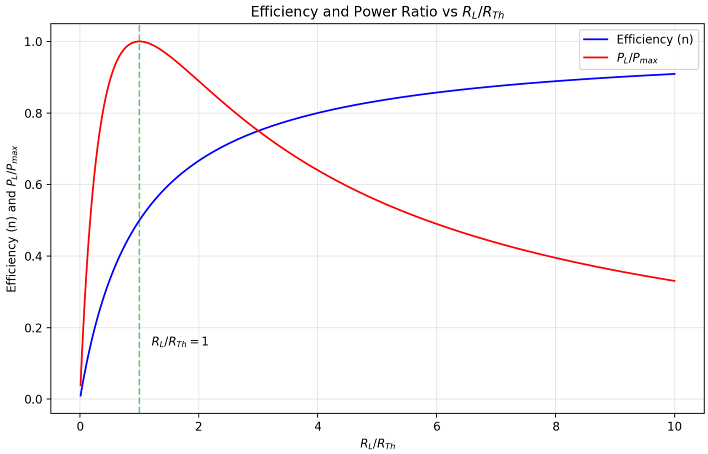


Figure 1: The red curve shows the power in the load, normalized relative to its maximum possible. The blue curve shows the efficiency η.

Linearity in Electrical Systems

A system is considered linear if it satisfies the properties of **Homogeneity** and **Additivity**.

Homogeneity

A system is homogeneous if the output is directly proportional to the input. In mathematical terms:

$$\text{If}\, x(t) \to y(t),\, \text{then}\, \alpha \cdot x(t) \to \alpha \cdot y(t)$$

where α is a scalar constant.

Additivity

A system is additive if the output to a sum of inputs is the sum of the outputs to each input. In mathematical terms:

$$\text{If}\, x_1(t) \to y_1(t)\, \text{and}\, x_2(t) \to y_2(t),\, \text{then}\, (x_1 + x_2)(t) \to y_1(t) + y_2(t)$$

Linear Elements: R, L, and C

The elements R (Resistor), L (Inductor), and C (Capacitor) are considered linear because they satisfy both homogeneity and additivity.

Resistor (R)

A resistor follows Ohm's Law:

$$V = IR \tag{1}$$

Applying the **homogeneity property**, if the current is scaled by a factor α,

$$I' = \alpha I \quad \Rightarrow \quad V' = (\alpha I)R = \alpha V. \tag{2}$$

For the **additivity property**, if two currents I_1 and I_2 pass through the resistor, their combined effect is:

$$I_{\text{total}} = I_1 + I_2 \quad \Rightarrow \quad V_{\text{total}} = (I_1 + I_2)R = V_1 + V_2. \tag{3}$$

Since both properties hold, a resistor is a **linear** element.

Inductor (L)

The voltage-current relationship for an inductor is:

$$V = L\frac{dI}{dt} \tag{4}$$

By the **homogeneity property**, scaling I by α results in:

$$I' = \alpha I \quad \Rightarrow \quad V' = L\frac{d(\alpha I)}{dt} = \alpha L\frac{dI}{dt} = \alpha V. \tag{5}$$

For **additivity**, if two currents I_1 and I_2 flow through the inductor:

$$I_{\text{total}} = I_1 + I_2 \quad \Rightarrow \quad V_{\text{total}} = L\left(\frac{dI_1}{dt} + \frac{dI_2}{dt}\right) = V_1 + V_2. \tag{6}$$

Thus, an inductor is a **linear** element.

Capacitor (C)

The current-voltage relationship for a capacitor is:

$$I = C\frac{dV}{dt} \tag{7}$$

By the **homogeneity property**, scaling V by α gives:

$$V' = \alpha V \quad \Rightarrow \quad I' = C\frac{d(\alpha V)}{dt} = \alpha C\frac{dV}{dt} = \alpha I. \tag{8}$$

For **additivity**, if two voltages V_1 and V_2 are applied:

$$V_{\text{total}} = V_1 + V_2 \quad \Rightarrow \quad I_{\text{total}} = C\left(\frac{dV_1}{dt} + \frac{dV_2}{dt}\right) = I_1 + I_2. \tag{9}$$

Since both properties hold, a capacitor is a **linear** element.

Conclusion: The elements R, L, and C are all linear, as they satisfy both homogeneity and additivity.

Active and Passive Elements

Active Elements

Active elements can supply energy to a circuit, such as a current or voltage source.

Passive Elements

Passive elements do not supply energy and include resistors, inductors, and capacitors.

Natural Response

Introduction

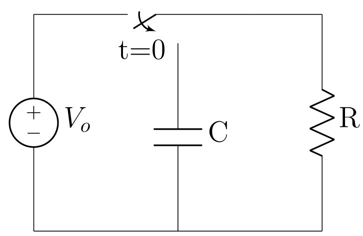
The behavior of electrical circuits and mechanical systems depends on their energy sources. When an external energy source determines the system's behavior, it is called a **forced response**. On the other hand, when the behavior is due to the system's internal energy storage, it is called a **natural response**.

For example, in an automobile, the driver controls speed and direction using the engine's energy. However, if the car is taken out of gear while moving, it will gradually slow down and stop due to internal energy dissipation. This slowing down happens in a predictable way, without any external force, illustrating a natural response.

A forced response can continue indefinitely as long as energy is supplied. In contrast, a natural response fades over time due to unavoidable energy loss, such as electrical resistance in circuits or friction in mechanical systems. When the natural response becomes negligible, the system reaches a **steady state**. The time between the start of the natural response and when it dies out is called the **transient period**.

In an electrical circuit, the external energy sources are usually voltage sources or current sources. Meanwhile, energy can be stored internally in the electric field of a capacitor or in the magnetic field of an inductor. These stored energy sources influence how the circuit behaves over time.

Capacitor's natural response



General steps to find the natural response:

1. Write governing equations (KCL/KVL) for $t \geq 0^+$:

$$-V_c + iR = 0$$

2. Reduce the governing equation to a homogeneous equation and differentiate:

$$-\frac{dV_c}{dt} + R\frac{di}{dt} = 0$$

   Homogeneous first-order equation:

$$\frac{i}{C} + R\frac{di}{dt} = 0$$

3. Assume an exponential solution template:

$$i = Ae^{st}$$

4. Plug the template solution into the homogeneous equation and solve for constants:

$$Ae^{st} + RAse^{st} = 0$$

   Solving for s:

$$s = -\frac{1}{RC}$$

   Hence, the current: $i = Ae^{-t/RC}$

5. To find A, use initial conditions. The default initial conditions are:
   - Capacitors: $V(t = 0^+) = V(t = 0^-)$
   - Inductors: $i(t = 0^+) = i(t = 0^-)$
6. At $t = 0^-$, voltage across the capacitor was V_0, so at $t = 0^+$, voltage across the capacitor remains $V_c = V_0$.
7. At $t = 0^+$, the current should be:
$$i = \frac{V_0}{R}$$
   Substituting into the general solution:
$$\frac{V_0}{R} = Ae^0$$
$$A = \frac{V_0}{R}$$
8. Hence, the final solution is:
$$i = \frac{V_0}{R}e^{-t/RC}, \quad t \geq 0$$
$$V_c = V_0 e^{-t/RC}, \quad t \geq 0$$
$$i = \frac{V_e^{-t/RC}}{R}$$

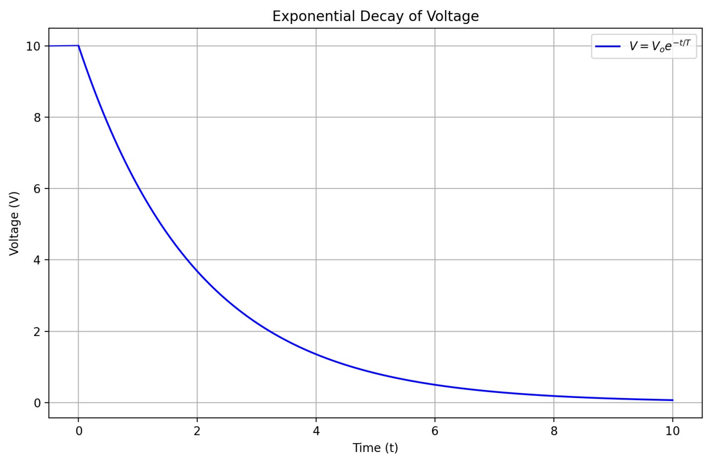


Figure 2: The graph shows the variation of Voltage with time with V_0 as 10V and τ as 2s

Time Constant

The time constant is given by:
$$T = RC$$
For $t = T$:
$$V = V_0 e^{-t/T}$$
Larger the time constant (T), slower the discharge.

Time	Voltage
0	V_0
T	$V_0/e = 0.37V_0$
$2T$	$V_o/e^2 = 0.14V_0$
$3T$	$0.05V_0$
$4T$	$0.018V_0$

Table 1: Variation of voltage with time in a RC circuit

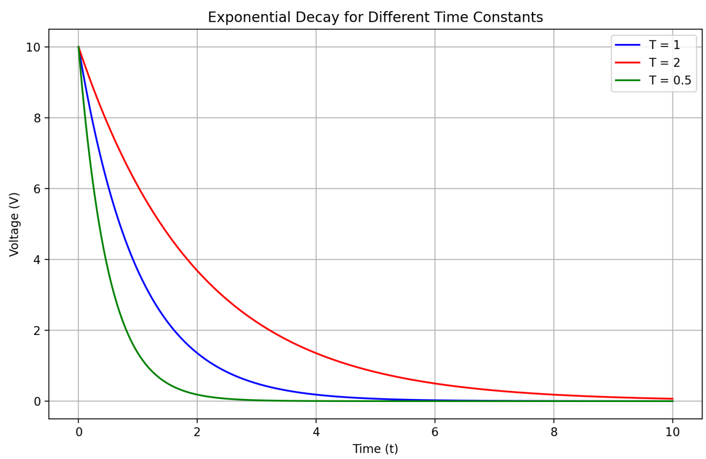
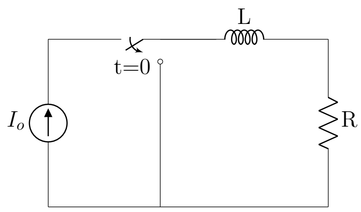


Figure 3: The graph shows the variation of Voltage with τ

$$\tau_3 > \tau_2 > \tau_1$$

Inductor's natural response



Applying Kirchhoff's Voltage Law (KVL):

$$V_L + iR = 0$$

$$L\frac{di}{dt} + iR = 0$$

Using the template solution $i = Ae^{st}$:

$$LAse^{st} + Ae^{st}R = 0$$

$$LsAe^{st} + Ae^{st}R = 0$$

Factoring:

$$Ae^{st}(Ls + R) = 0$$

For a non-trivial solution, we set:

$$Ls + R = 0$$

Solving for s:

$$s = -\frac{R}{L}$$

Applying initial condition $i(0^+) = I_0$:

$$A = I_0$$

Thus, the final solution is:

$$i = I_0 e^{-t/(L/R)}$$

The time constant is:

$$T = \frac{L}{R}$$

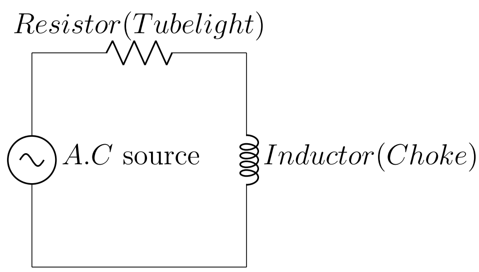
RC and RL Circuits

RC Circuit

A common usage of an RC circuit is in a camera. RC circuits are also used as timers in windshield wipers. We can increase or decrease the time delay by changing the time constant $\tau = RC$, i.e., adjusting R or C.

RL Circuit

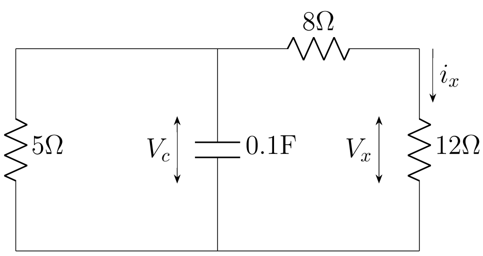
RL circuits are used in tube lights as chokes. Normally, an inductor is used here.



An inductor opposes the flow of current initially, so if the inductor is not present in the circuit shown, there will be a rapid increase in current because of the resistor, and a high current surge can damage the filaments inside the tube.

Q

Given the circuit below: - i_x: Current through the $12\,\Omega$ resistor. - v_x: Voltage across the $12\,\Omega$ resistor. - If $v_C(0) = 15\,\text{V}$, find v_C, v_x, i_x for $t \geq 0$.



Answer:
We know for an RC circuit:

$$V_C = V_0 e^{-t/\tau}$$

The equivalent resistance is given by:

$$R_{eq} = \frac{(8+12)\times 5}{8+12+5} = \frac{20\times 5}{25} = 4\,\Omega$$

Given that:

$$V_0 = 15V$$

Since $e^0 = 1$, we get:

$$V_C = 15e^{-t/\tau}$$

Substituting $\tau = R_{eq}C$ where $R_{eq} = 4$ and $C = 0.1$:

$$\tau = 4 \times 0.1 = 0.4$$

Thus,

$$V_C = 15e^{-t/0.4}$$

For $t = 1$:

$$V_C = 15e^{-2.5t}$$

To find V_x, we can first use the voltage divider:

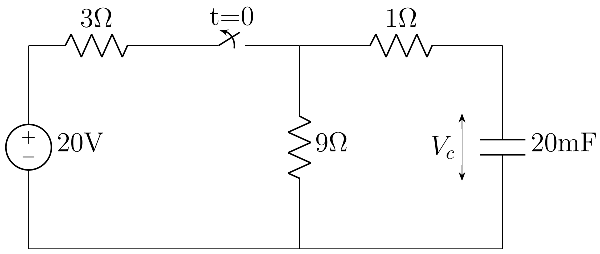
$$V_x = \frac{12}{12+8} \times V_C = \frac{12}{20} \times 15e^{-2.5t}$$

$$V_x = 9e^{-2.5t}$$

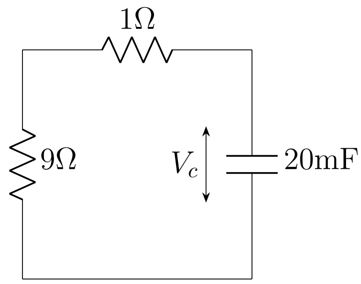
$$i_x = \frac{V_x}{12} = \frac{3}{4}e^{-2.5t}$$

Q

Assume the circuit was closed for a long time and opened at $t \geq 0$. Find V_c for $t \geq 0$.



Answer:
For $t \geq 0$:



$$V_c = V_0 e^{-t/RC}$$

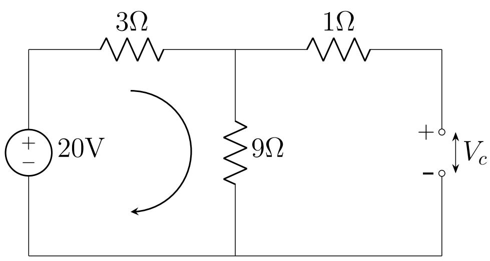
$$V_c = 10e^{-t/(10 \times 20 \times 10^{-3})}$$

$$V_c = V_0 e^{-5t}$$

To find V_0, let's analyze the circuit:

$$V_c(0^+) = V_c(0^-)$$

During the steady-state when there is a D.C source, no current flows through the capacitor.



$$V_o = V_c = \frac{9 \times 20}{9+3} = 15V$$
$$t(0^+) = t(0^-)$$

Remember: This is the table showing behaviour of components from $t = (0^-)$ to $t = (0^+)$:

Component	**Property**
Inductor	$V(0^-) = V(0^+)$
Capacitor	$I(0^-) = I(0^+)$
Resistor	N.A

Table 2: Component Behaviour Table

Remember: This is the table showing behaviour of components at t=∞

Component	**Property**
Inductor	Short circuit
Capacitor	Open circuit
Resistor	No change

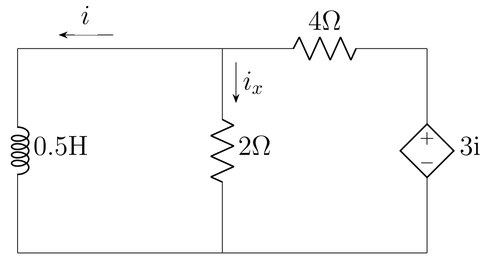
Table 3: Component Behaviour Table

So...

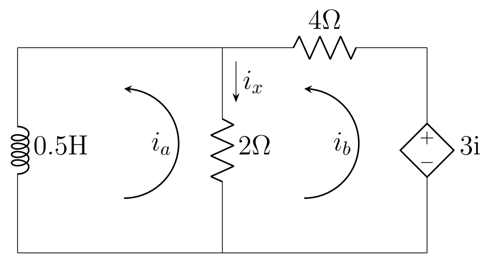
$$V_C = V_0 e^{-t/\tau}$$
$$V_C = 15e^{-5t}$$

Q

Assume i(0)=10 A. Find i and i_x.



Answer:



So by writing KVL,

$$a\theta V_L + 2(i_a - i_b) = 0$$

$$L\frac{di_a}{dt} + 2(i_a - i_b) = 0$$

For loop b,

$$2(i_b - i_a) \text{ or } 3i_a + 4i_b = 0$$

And i = i_a, so

$$6i_b - 5i_a = 0$$

$$6i_b = 5i_a$$

$$i_b = \frac{5i_a}{6}$$

Substituting into the first equation,

$$L\frac{di_a}{dt} + 2 \times \left(i_a - \frac{5i_a}{6}\right) = 0$$

$$L\frac{di_a}{dt} + \frac{i_a}{3} = 0$$

$$\frac{di_a}{dt} = -\frac{2i_a}{3}$$

Our template solution is:

$$i_a = Ae^{st}$$

Given that:

$$i_a = Ae^{st}$$

Taking the derivative:

$$\frac{di_a}{dt} = Ase^{st}$$

Substituting into the given equation:

$$Ase^{st} = -\frac{2}{3}Ae^{st}$$

$$Ae^{st}\left(s + \frac{2}{3}\right) = 0$$

For a non-trivial solution, we get:

$$s + \frac{2}{3} = 0$$

Solving for s:

$$s = -\frac{2}{3}$$

Thus, the general solution is:

$$i_a = Ae^{-2t/3}$$

Using the initial condition i(0) = 10:

$$10 = Ae^0$$

$$A = 10$$

Thus, the final solution is:

$$i_a = 10e^{-2t/3}$$

$$i$$

$$i_x = i_b - i_a$$

$$i_x = \frac{5}{6}i_a - i_a$$

$$i_x = \frac{-1}{6}i_a$$

$$i_x = \frac{-1 \times 10e^{-2t/3}}{6}$$

$$i_x = \frac{-5e^{-2t/3}}{3}$$

Contribution-

Ehsan- Lecture 4 notes and fully formatting it in overleaf

Farhan- Lecture 5 notes and formatting it in overleaf

Himanshu - Lecture 6 notes and formatting it in overleaf

Abhiram - Drawing circuits and graphs and overall supervision on the notes